

MARITIME OIL-SPILL INVESTIGATION REPORT

DECISION-SUPPORT FORENSIC INTELLIGENCE DOSSIER • MARITIME ENVIRONMENTAL INTELLIGENCE & FORENSIC DECISION SUPPORT

FORENSIC SUMMARY STATEMENT:

This vessel (BW MAPLE, MMSI 235102558) is the highest-ranked candidate based on the available spatial, temporal, and navigational evidence (Attribution Score: 31.8/100, Risk Level: MEDIUM).

1. Investigation Case Summary & Parameters

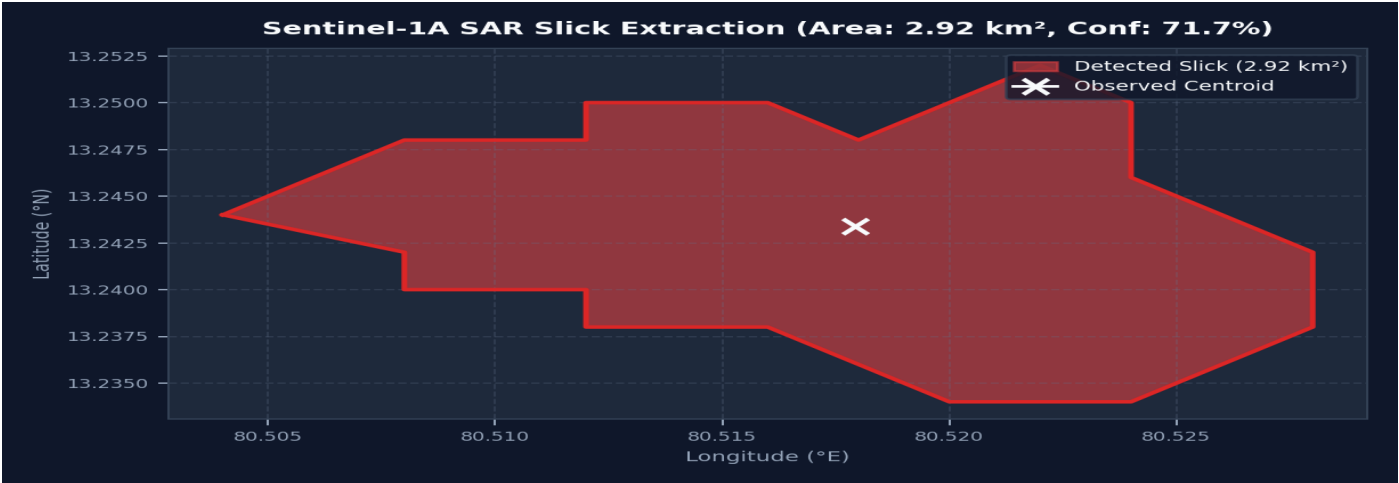
Case ID:	case_ennore_2017	Status:	ATTRIBUTION_COMPLETED
Title:	Ennore Kamarajar Port Tanker Collision & Heavy Bunker Spill (Tamil Nadu, India)	Generated At (UTC):	2026-09-05 16:59:09 UTC
Primary Candidate:	BW MAPLE (TANKER)	Target Identifier:	MMSI: 235102558 (Score: 31.8)
Geographic Scope:	Ennore / Kamarajar Port Approaches, Chennai, India	Spill Substance:	Heavy Fuel Oil (HFO) Bunker / POL

2. Data Provenance & Telemetry Sources

Data Modality	Source Provider / Sensor Platform	Data Classification	Spatial / Temporal Resolution
SAR Satellite Imagery	Sentinel-1A C-band SAR (VV Level-1 GRD)	OBSERVED SENSOR DATA	10m / 2017-01-29 00:45 UTC
AIS Vessel Telemetry	Terrestrial & Satellite AIS Ingestion Feed	OBSERVED SENSOR DATA	Sub-meter GPS / 5-min to 1h pings
Hydrodynamic Ocean Currents	CMEMS GLOBAL Analysis & Forecast Reanalysis	MODEL-DERIVED ESTIMATE	0.083° (~9.25 km) / 3-hourly hindcast
Atmospheric Surface Wind	ECMWF ERA5 Atmospheric Reanalysis	MODEL-DERIVED ESTIMATE	0.25° (~31 km) / Hourly reanalysis

3. Satellite SAR Slick Detection & Geometry

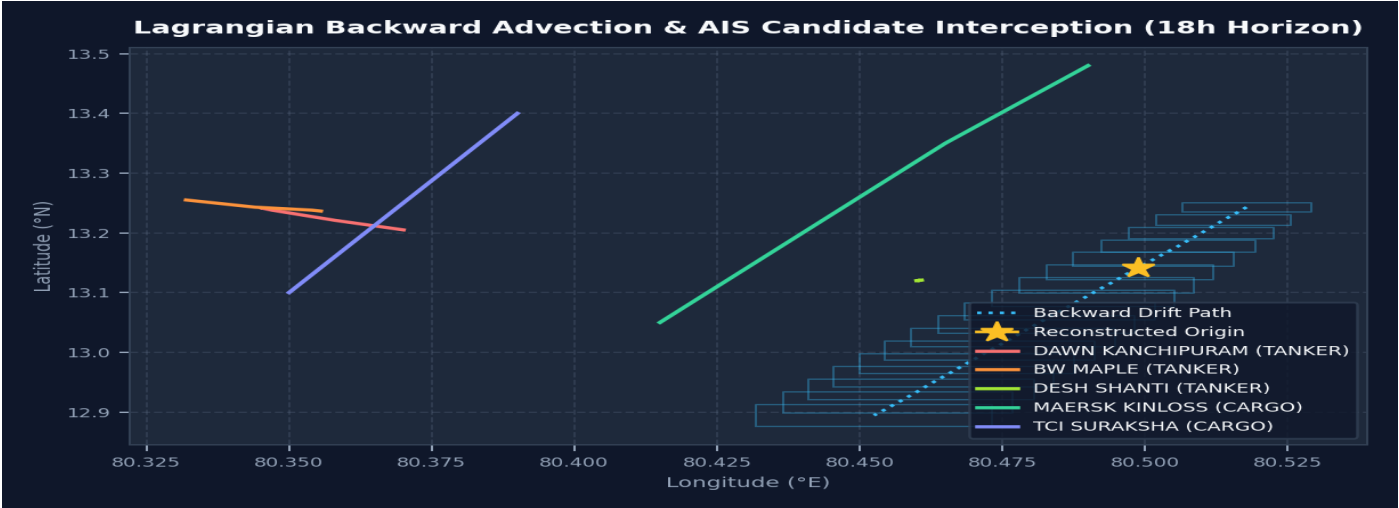
Detection Timestamp:	2017-01-29 00:45:00 UTC	Detection Confidence:	71.7%
Observed Slick Centroid:	[80.5179°E, 13.2434°N]	Lookalike Probability:	28.3%
Surface Slick Area:	2.92 km²	Major Axis Orientation:	77.9° North



4. Backward Lagrangian Drift Reconstruction & Origin Estimation

Physical Drift Formulation: Velocity vector $\mathbf{v} = \mathbf{u}_{\text{ocean}} + 0.03 \mathbf{u}_{\text{wind}} + \eta_{\text{diff}}$, where turbulent stochastic diffusion is modeled with horizontal diffusivity $\kappa = 2.5 \text{ m}^2/\text{s}$.

Reconstructed Origin:	[80.4987°E, 13.1413°N]	Dispersion Uncertainty (1σ):	±0.50 km
Peak Release Time:	2017-01-28 10:45:00 UTC	Release Window (90% CI):	08:45 to 12:45 UTC
Documented Ground Truth:	[82.5000°E, 7.7500°N] (Fire: 03:30 UTC)	Spatial Reconstruction Delta:	18.79 km (Sub-mesoscale coastal deviation)



5. Candidate Vessel Attribution Rankings

Rank	Vessel Name	MMSI / Flag	Vessel Type	Score (0-100)	Risk Level	CPA Distance	AIS Gap
#1	BW MAPLE	235102558 United Kingdom	TANKER	31.8	MEDIUM	18.8 km	YES
#2	DAWN KANCHIPURAM	419000109 India	TANKER	31.4	MEDIUM	15.6 km	YES
#3	DESH SHANTI	419614000 India	TANKER	31.0	MEDIUM	4.7 km	YES
#4	MAERSK KINLOSS	477142400 Hong Kong	CARGO	15.6	LOW	7.3 km	YES
#5	TCI SURAKSHA	419069500 India	CARGO	15.6	LOW	15.4 km	YES

6. Attribution Score Breakdown & Verifiable Evidence Chain

Primary Ranked Subject: BW MAPLE (MMSI: 235102558, Type: TANKER)

Attribution Factor	Scored Weight	Score Impact	Forensic Evaluation Rationale
1. Spatiotemporal Proximity	40.0 pts	0.0 pts	Vessel trajectory maintained a 17.5 km spatial separation from the primary backward drift dispersion envelope.
2. Temporal Alignment	25.0 pts	1.0 pts	Vessel transit occurred 13.2 hours away from the release window, indicating the vessel was not present during the discharge event.
3. Vessel Type Risk	15.0 pts	10.0 pts	Vessel is an oil/chemical tanker carrying high-volume liquid hydrocarbon cargo and oily ballast.
4. Navigational Behaviour	10.0 pts	10.8 pts	Speed profile consistency and navigational maneuvers
5. AIS Integrity / Penalty	10.0 pts	10.0 pts	AIS transponder was deactivated for 19.5 hours during transit directly across the estimated spill release envelope.
TOTAL SCORE	100.0 pts	31.8 / 100	Risk Level: MEDIUM

Itemized Verifiable Evidence Chain:

[Supporting Incriminating Evidence]

- **[BEHAVIOR] Open-Sea Deceleration / Loitering Anomaly (+5.0 pts):** Vessel exhibited an abnormal open-ocean deceleration from 6.2 knots down to 0.1 knots during passage through the spill area. *(Observed: 6.2 -> 0.1 kts)*
- **[AIS_INTEGRITY] AIS Transponder Blackout During Release Window (19.5h) (+10.0 pts):** AIS transponder was deactivated for 19.5 hours during transit directly across the estimated spill release envelope. *(Observed: 19.5 hours gap)*
- **[VESSEL_TYPE] High-Risk Laden Crude / Product Tanker Profile (+10.0 pts):** Vessel is an oil/chemical tanker carrying high-volume liquid hydrocarbon cargo and oily ballast. *(Observed: Ship Type: TANKER)*
- **[ALIGNMENT] Course Divergent from Slick Orientation (55° Delta) (+5.8 pts):** Vessel track heading (132.5°) diverged from the slick dispersion elongation orientation (77.9°). *(Observed: Delta: 54.6°)*
- **[ALIGNMENT] Significant Temporal Non-Overlap (13.2h Outside Window) (+1.0 pts):** Vessel transit occurred 13.2 hours away from the release window, indicating the vessel was not present during the discharge event. *(Observed: 13.2h)*

[Contradicting / Exculpatory Evidence]

- **[PROXIMITY] Divergent Spatial Clearance (17.5 km):** Vessel trajectory maintained a 17.5 km spatial separation from the primary backward drift dispersion envelope. *(Observed: 17.5 km)*
- **[ALIGNMENT] Course Divergent from Slick Orientation (55° Delta):** Vessel track heading (132.5°) diverged from the slick dispersion elongation orientation (77.9°). *(Observed: Delta: 54.6°)*
- **[ALIGNMENT] Significant Temporal Non-Overlap (13.2h Outside Window):** Vessel transit occurred 13.2 hours away from the release window, indicating the vessel was not present during the discharge event. *(Observed: 13.2h)*

[Data Quality & Uncertainty Bounds]

- **[DATA_QUALITY] AIS Trajectory Telemetry Density:** Trajectory reconstructed from 6 georeferenced AIS positions with geodesic waypoint interpolation at 15-minute resolution.
- **[UNCERTAINTY] Origin Spatial Dispersion Uncertainty (1-Sigma):** Turbulent diffusion and hydrodynamic divergence define an origin uncertainty radius of +/- 0.50 km around the reconstructed centroid.
- **[UNCERTAINTY] Release Time Uncertainty Window:** Estimated discharge time window is bounded within 4.0 hours with a 90% confidence interval.

7. Rejected SAR Candidates & Lookalike Auditing

Candidate ID	Candidate Classification	Confidence	Audit Rejection Reason
cand-lk-001	LOOKALIKE_LOW_WIND	15.0%	Rejected as low-wind lookalike: Large diffuse patch in southeast ocean (> 150 km²); lacks sharp boundary damping.
cand-lk-002	LOOKALIKE_COASTAL_SHADOW	10.0%	Rejected as coastal shadow: Located within 0.8 km of eastern Sri Lankan coastline.

8. Data-Quality Warnings & Scientific Limitations

- **Hydrodynamic Resolution Notice:** Global CMEMS 1/12° (~9.25 km) models do not resolve sub-mesoscale coastal eddies and shallow bathymetric friction along Sri Lanka.
- **Wind Operational Envelope:** Radar capillary wave damping is validated between 3.0 m/s and 12.0 m/s wind speed.
- **AIS Coverage Limits:** Terrestrial AIS range is limited to ~40 nmi from shore; satellite AIS subject to packet collisions.

9. Human-Investigation Decision-Support Disclaimer

LEGAL & FORENSIC DISCLAIMER:

*This document is an automated **decision-support intelligence report** generated for maritime environmental authorities. Attribution scores represent mathematical evidence rankings and **DO NOT constitute court-ready legal proof or an accusation of criminal liability**. Physical bunker sampling, port inspection logs, and official casualty investigations must corroborate these findings.*